1.0.0 PLUMBING FIXTURES

Plumbing fixtures include faucets, tanks, and receptacles in kitchens and bathrooms. There are many types and styles of fixtures; some are general, while others have been adapted to meet special applications, such as for hospitals, prisons, and similar institutions. Military installations usually are planned to house large numbers of personnel, and the plumbing fixtures ordinarily are installed in batteries. Instructions for installing fixtures are given either by the manufacturer or by specifications. Sometimes you may have to design and lay out a fixture or battery of fixtures. You must know what water supplies and stack sizes are needed and work these into your design.

Standard plumbing fixtures are individually tested so that the amount of liquid waste that can be discharged through their outlet ***orifices*** in a given interval is measured. The fixture unit value for different plumbing fixtures is shown in *Table 6-1*. The basis for the fixture unit system comes from the fact that the washbasin, one of the smaller fixtures discharges 1 cubic foot of water per minute.

Table 6-1 — Plumbing Fixture Unit Values.

FIXTURE	UNITS
Lavatory or washbasin	1
Kitchen sink	2
Bathtub	2
Laundry tub	2
Combination fixture	3
Urinal	5
Shower bath	2
Floor drain	1
Slop sink	3
Water closet	6
180 square feet of roof drained	1

Plumbing fixtures must be furnished with water at a rate of flow that will fill it within a reasonable time, as well as, with a waste pipe of sufficient capacity to carry off all water supplied to it quickly and quietly.

Table 6-2 shows the minimum supply pipe diameters, according to the *National Standard Plumbing Code*

Table 6-2 — Minimum Size Fixture Supply.

FIXTURE	Supply Pipe Diameter Min. Size
Water closet (tank type)	1/2 inch
Water closet	1 inch
Flushometer urinal with flushing valve	3/4 inch
Laundry tubs	1/2 inch
Kitchen sink	1/2 inch
Lavatory	1/2 inch
Slop sink	1/2 inch
Drinking fountain	1/2 inch
Shower	1/2 inch

1.1.0 Rough-In Measurements

Figure 6-1 shows general rough-in measurements for a tank-type water closet. These measurements vary depending on the type of fixture and the manufacturer. It is your responsibility to identify the fixtures you will be using so you can obtain the proper rough-in measurements.

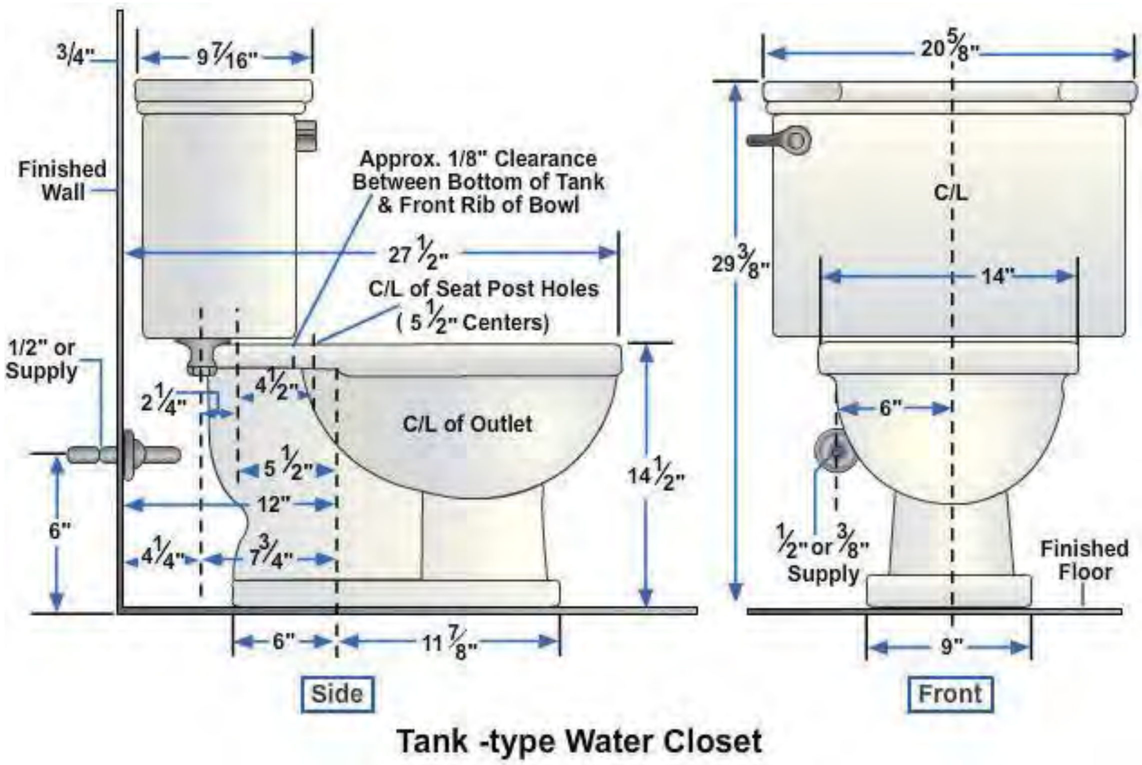

Figure 6-1 - Rough-in measurements for a tank-type water closet

Service connections for steam radiators depend upon their sizes and location. The same holds true for water tanks used for storing or heating.

After roughing-in, you can easily install the plumbing fixtures and trim work. Instructions are given here for the installation of various fixtures and accessories. Even though not every type of fixture is included. if you learn to install the fixtures covered in *Figure 6-1*, you should not have any problems with other types.

1.2.0 Water Closets

Water closets, devices designed to receive human waste and dispose of it properly in a sanitary sewer system, come in various shapes, designs, and colors. Most water closets mount on the floor, but some models are wall-hung. Modem water closets have various design features which create different flushing actions.

1.2.1 Installation

To install a water closet, shown in *Figure 6-2*, follow these procedures as a general guide.

1. Slip the water closet flange over the closet bend and slide it down until it is level with the finish floor.
2. With a hammer and cold chisel, break off the portion of the closet bend that projects above the water closet flange. Do not break the closet bend below the flange.
3. Place the two brass closet hold-down bolt heads in the slots of the flange.
4. On the bottom of the water closet, as shown in *Figure 6-3, View A*, slip the preformed sealing ring over the horn to form a sealing gasket for the water closet against the face of the flange. Do not use putty as it will dry out and leave a possible sewer gas leak.
5. Turn the water closet bowl right side up and set it on the flange with the horn projecting down into the flange. Place a wedge under the low side of the water closet if unlevel. In setting the bowl on the flange, as shown in *Figure 6-3, View B*, guide the two hold-down bolts up through the bolt holes on either side of the base of the water closet. Use your full weight to press down and twist

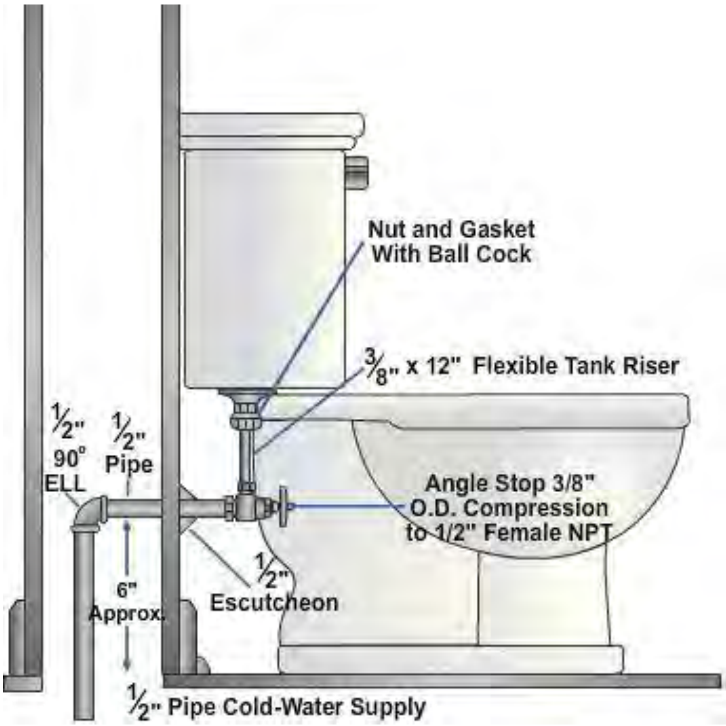

Figure 6-2 — Water closet (tank type).

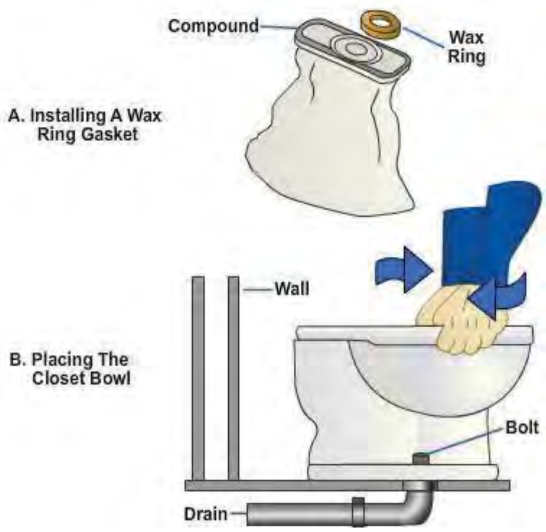

Figure 6-3 — Setting a closet bowl.

slightly to settle the bowl and the wax ring into position. The bowl should be perfectly level when settled.

6. Install nuts on the hold-down bolts and tighten them alternately. Do not over tighten them as this may crack the base of the water closet.

7. A wall-mounted water closet is attached to the wall by a chair carrier, similar to the one shown in *Figure 6-4.* The chair carrier is positioned and bolted to the floor. A standard fitting is used to connect between the drain and the closet bowl after the chair carrier is bolted down. The fittings are for 4-inch iron or soil pipe. The bolt holes in the chair carrier are slotted to facilitate installation of the closet bowl.

When mounting a close-coupled tank on a closet bowl, note that two bolts hold the tank on the bowl, as *Figure 6-5* illustrates. The water supply pipe is between the bolts and drops the water directly into the bowl. A specially designed gasket is installed between the tank and bowl to make the connection waterproof. The bolts are tightened from underneath the closet bowl. Do not apply too much pressure when you tighten these bolts, because you may crack the bottom of the tank or the back of the bowl.

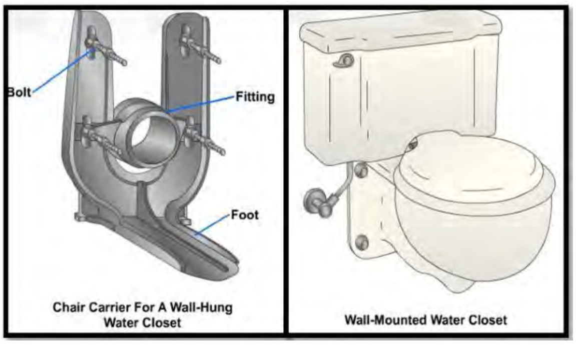

Figure 6-4 – Wall-mounted water closet.

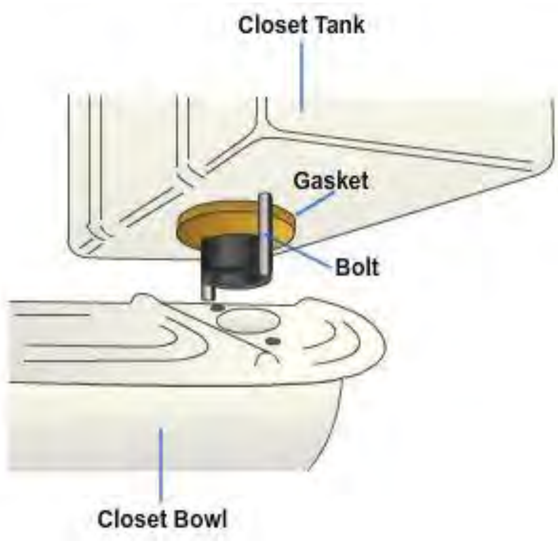

Figure 6-5 – Mounting a close coupled tank.

After the tank is firmly attached to the bowl, connect the water supply pipe to the tank inlet with a riser tube, as shown in *Figure 6-6*. The jiffy connector used here is the same as the connector used to connect the water supply to the faucets of a lavatory. The rubber washer shown in *Figure 6-6* is commonly referred to as a doughnut washer because of its thickness.

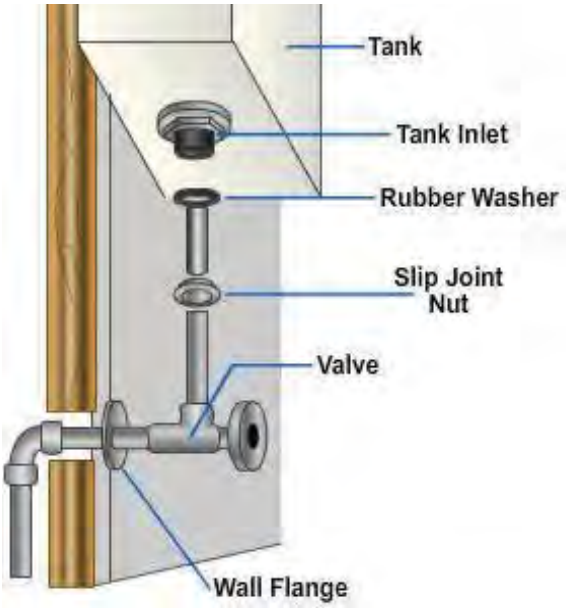

Figure 6-6 — Closet tank water supply line.

1.2.2 Flushometer Valves

Because of how they operate, flushometers are referred to in two categories, piston and diaphragm. In some applications they are used in place of a tank-type valve. When a flush valve is used, no tank to hold the flushing mechanism or water is required, therefore, larger pipe diameters are required.

Flush valves require less water volume per flush than a tank-type valve, provide quicker multiple flushing capability, and lower maintenance costs in commercial applications; however, they are noisier, initially cost more, and require a higher operating pressure than tank-type valves. These items usually require larger pipe sizes. Basically, the flush valve is suited more for the commercial or industrial application, and the flush tank is used in small buildings or single family dwellings.

A backflow preventer or vacuum breaker, such as the type shown in *Figure 6-7*, should be installed on the discharge side of a flushometer and on the supply line of a float valve in a water closet tank, if the tank outlet is below the flood level rim of the closet bowl.

Figure 6-7 – Flushometer valve showing a backflow preventer.

A detail of a diaphragm type of flushing valve is shown in *Figure 6-8*. The diaphragm type of valve consists of an upper and lower chamber. These chambers are separated by the diaphragm and relief valve. The lower chamber is connected directly to the incoming water supply. This incoming water is the flushing water and also shuts the valve off after the flushing period. The valve is flushed by the diaphragm and the water pressure in the upper chamber. Water is forced into the upper chamber through a small orifice (hole) in the diaphragm. Water pressure, passing through this orifice into the upper chamber, creates the pressure required to force the diaphragm down and shuts off the flushing water. When the flushing handle is moved, the relief valve tilts open. Then pressure decreases in the upper chamber to less than that of the incoming and flushing water. The action allows the flushing water pressure to raise the diaphragm off the flushing seat and recycle.

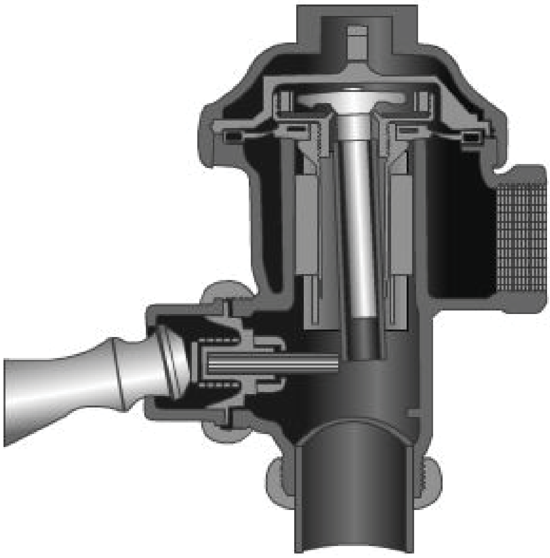

Figure 6-8 – Diaphragm type of flushing valve.

Figure 6-9 shows a piston type of valve. With this type of flush valve, the piston is drawn up when the flushing begins, then to its closed position by the filling of the upper chamber through the expeller orifice tube.

Flush valve assemblies on urinals and water closets may be protected from unnecessary damage and wear by installing a grip handle or guard firmly over the handle housing. This grip handle increases the operating life of flush valves and thereby reduces service calls on the repair of flush valve assemblies and plumbing fixtures.

Figure 6-9 – Piston type of flushing valve.

1.3.0 Urinals

Two major types of urinals in use are the floor-mounted and wall-mounted urinals. We will discuss some of the main items relating to the installation of the wall-hung urinal (*Figure 6-10*). Before installing a floor-mounted urinal, refer to local codes to ensure it is legal.